

HWK 4
ADVANCED MACHINE LEARNING
DATA 642

Objective: This homework aims to deepen students' understanding of Principal Component Analysis (PCA) and Independent Component Analysis (ICA) through theoretical derivations and practical applications with real-world audio data. In the first exercise, students will theoretically demonstrate that the second principal component in PCA is given by the eigenvector corresponding to the second largest eigenvalue. The second exercise involves a more complex scenario where students will simulate a cocktail party problem with mixed audio signals from three sources: two voices and one music track. Using the provided audio files, students will generate mixed signals by applying a random mixing matrix, and then use FastICA to recover the original sources. They will also compare the separation performance of FastICA with PCA and the ICA-EBM algorithm. Through these exercises, students will gain hands-on experience with signal processing techniques and enhance their ability to analyze and interpret the effectiveness of different decomposition methods in separating mixed signals.

Exercise 1 (5 points)

Show that the second principal component in PCA is given as the eigenvector corresponding to the second largest eigenvalue.

Exercise 2 (10 points)

In a party, there are various people speaking; in our case, we are going to consider music as well. Let us say that there are people (a female and a male) and there is also monophonic music, making three sources of sound in total. Then, three microphones (as many as the sources) are placed in different places in the room and the mixed speech signals are recorded. In the simplest of the models, the three recorded signals can be considered as linear combinations of the individual source signals. Delays are not considered. The goal is to use ICA and recover the original speech and music from the recorded mixed signals.

To this end and in order to bring the task in the formulation we have previously adopted, we consider the values of the three signals at different time instants as different observations of the corresponding random variables, x_1 , x_2 , and x_3 , which are put together to form the random vector $\mathbf{x}$. We further adopt the very reasonable assumption that the original source signals, denoted as $s_1(t)$, $s_2(t)$, and $s_3(t)$, are statistical independent and the values at different time instants correspond to the values of three latent variables, denoted together as a random vector $\mathbf{s}$.

The two voice and the music signals can be downloaded from Canvas and read using the read function from the SoundFile (<https://pypi.org/project/soundfile/>) Python package. Generate a random mixing matrix $\mathbf{A}$, (3×3) and produce with it the three mixture signals. Each one of them simulates the signal received in each microphone. Then, apply FastICA in order to estimate the source signals. Use the play function from the sounddevice (<https://pypi.org/project/sounddevice/>) Python package in order to listen to the original signals, to the mixtures, and the recovered ones.

Repeat the previous steps performing PCA instead of ICA and compare the results. Repeat the experiment using the ICA-EBM algorithm. (<https://github.com/ibs-lab/cedalion/tree/main/src/cedalion/sigdecomp>). What do you observe in terms of the separation performance?